UFC FIGHT PROJECT

Group No.: 12

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GOALS

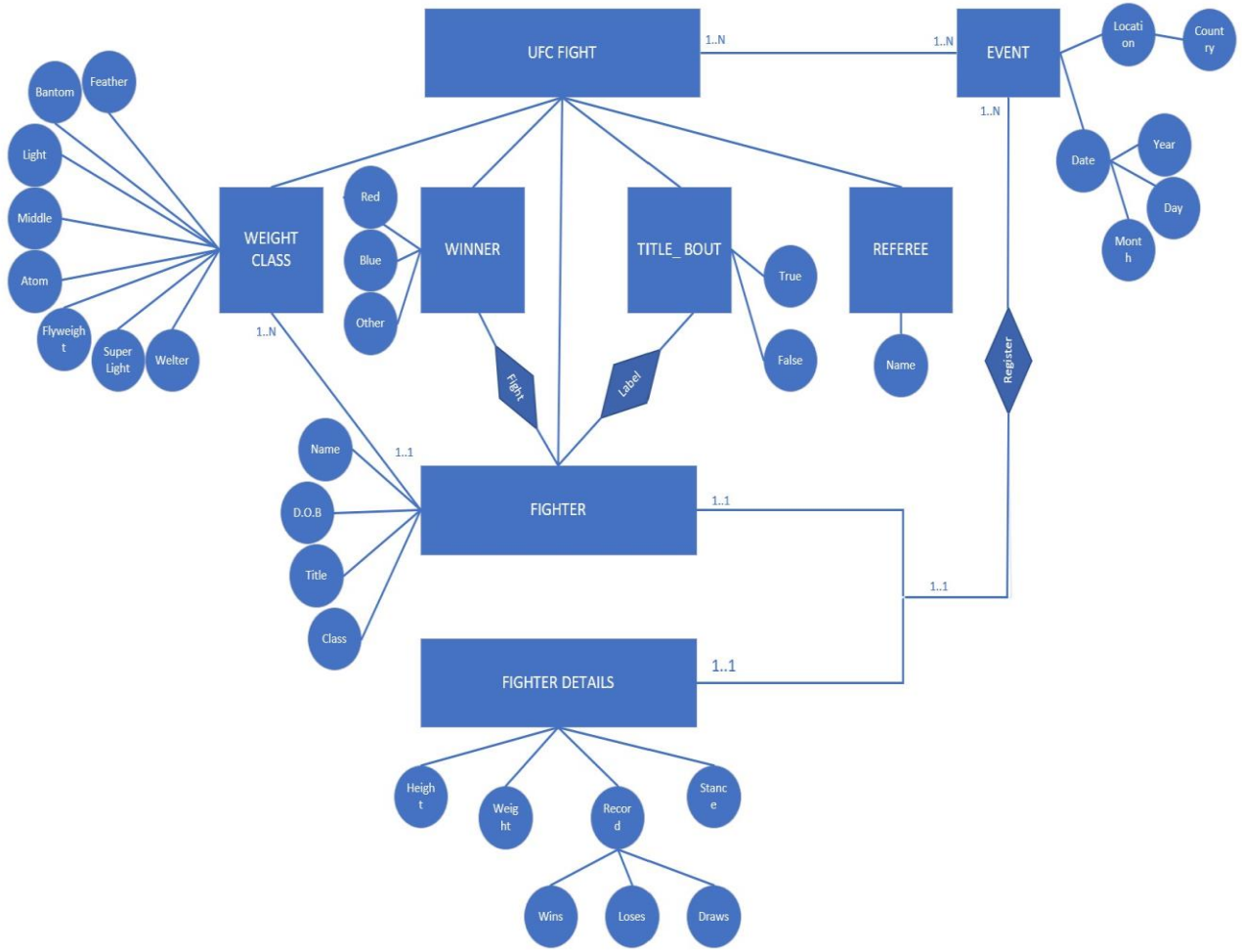
- My main research questions focus around looking at the evolution of fighting styles since the first UFC fight until 2021.

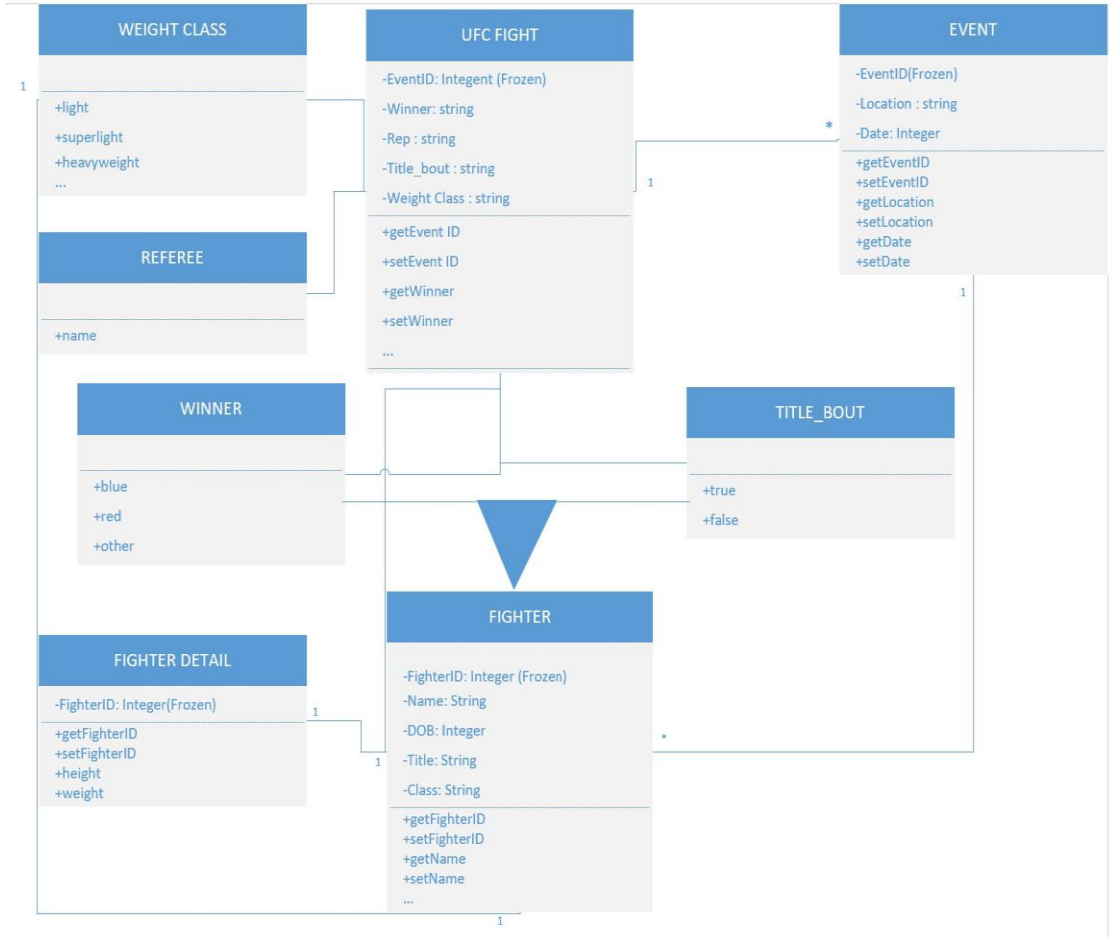
- I will dig deep into the type of data recorded regarding individual fighters, on a larger scale derive how it benefits the fighters in preparation for their upcoming bouts and of course, their coaches so they’re able to provide the best game plan and fitness recovery possible.

- I am using a dataset found on Kaggle.com. The dataset contains historical fight data from 1993 until March of 2021.

https://www.kaggle.com/rajeevw/ufcdata/data#

CONCEPTUAL DATA MODEL





TOOLS

- Cleaned and organized raw data.
- Used to creates a database for storing and manipulating data, defining the relationship of each table.

SQL QUERIES with MySQL

SCHEMAS

Filter objects

- adventureworks
- sys
- test
- ufc
  - Tables
    - event
    - fighter
    - fighter_details
    - ufc_fight
    - winner
  - Views
  - Stored Procedures
  - Functions

Administration Schemas

Information

Table: winner

Columns:
R_fighter text
B_fighter text

Limit to 500 rows

```
#7. Find the fighter into southpaw stance with weight over 200lbs
select fighter.Fighter_ID, fighter.fighter_name, fighter.DOB, fighter_details.weight_cla
from fighter
Inner JOIN fighter_details On fighter.Fighter_ID = fighter_details.Fighter_ID
where Stance = "Southpaw"
and Weight >= 200;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Fighter_ID	fighter_name	DOB	weight_class	Stance	Height	Weight	Reach
11	Scott Adams		Flyweight	Southpaw	6' 0""	225 lbs.	
90	Sam Alvey	6-May-86	Flyweight	Southpaw	6' 2""	205 lbs.	75""
91	Jimmy Ambriz	26-Jan-70	LightHeavyweight	Southpaw	6' 2""	315 lbs.	
100	Matt Andersen	29-May...	Featherweight	Southpaw	6' 2""	230 lbs.	
121	Josh Appelt	19-Jun-83	Bantamweight	Southpaw	5' 10""	255 lbs.	

Result 93 | Read Only

Output

```
#6. Join ufc_fight with event table while winner is blue based on asc location
select ufc_fight.Event_ID, ufc_fight.Referee, ufc_fight.Winner, ufc_fight.title_bout, ufc_fight.weight_class
from ufc_fight
Inner JOIN event On ufc_fight.Event_ID = event.Event_ID
where winner = "Blue"
order by location asc;

#7. Find the fighter into southpaw stance with weight over 200lbs
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Event_ID	Referee	Winner	title_bout	weight_class	date	location
83	Jason Herzog	Blue	FALSE	Middleweight	1/23/2021	Abu Dhabi, Abu Dhabi, United Arab Emirates
85	Keith Peterson	Blue	FALSE	WomenFlyweight	1/23/2021	Abu Dhabi, Abu Dhabi, United Arab Emirates
91	Herb Dean	Blue	FALSE	LightHeavyweight	1/23/2021	Abu Dhabi, Abu Dhabi, United Arab Emirates
82	Marc Goddard	Blue	FALSE	Lightweight	1/23/2021	Abu Dhabi, Abu Dhabi, United Arab Emirates
92	Lukasz Bosacki	Blue	FALSE	CatchWeight	1/23/2021	Abu Dhabi, Abu Dhabi, United Arab Emirates
97	Keith Peterson	Blue	FALSE	WomenFlyweight	1/20/2021	Abu Dhabi, Abu Dhabi, United Arab Emirates
107	Jason Herzog	Blue	FALSE	WomenBantamweight	1/16/2021	Abu Dhabi, Abu Dhabi, United Arab Emirates

NoSQL with MongoDB

```
> db.ufc_fight.find({$and:[{Winner:"Red"},{weight_class:"Lightweight"}]}).limit(5)
< { _id: ObjectId("638eb47e58e77d912eb65180"),
    Event_ID: 19,
    Referee: 'Chris Tognoni',
    Winner: 'Red',
    title_bout: 'FALSE',
    weight_class: 'Lightweight',
    B_avg_KD: '',
    B_avg_opp_KD: '',
    B_avg_SIG_STR_pct: '',
    B_avg_opp_SIG_STR_pct: '',
    B_avg_TD_pct: '',
    B_avg_opp_TD_pct: '',
    B_avg_SUB_ATT: '',
    B_avg_opp_SUB_ATT: '',
    B_avg_REV: '',
    B_avg_opp_REV: '',
    B_avg_SIG_STR_att: '',
    B_avg_SIG_STR_landed: '',
    B_avg_opp_SIG_STR_att: '',
    B_avg_opp_SIG_STR_landed: '',
    B_avg_TOTAL_STR_att: '',
    B_avg_TOTAL_STR_landed: '',
    B_avg_opp_TOTAL_STR_att: ''
```

{} My Queries

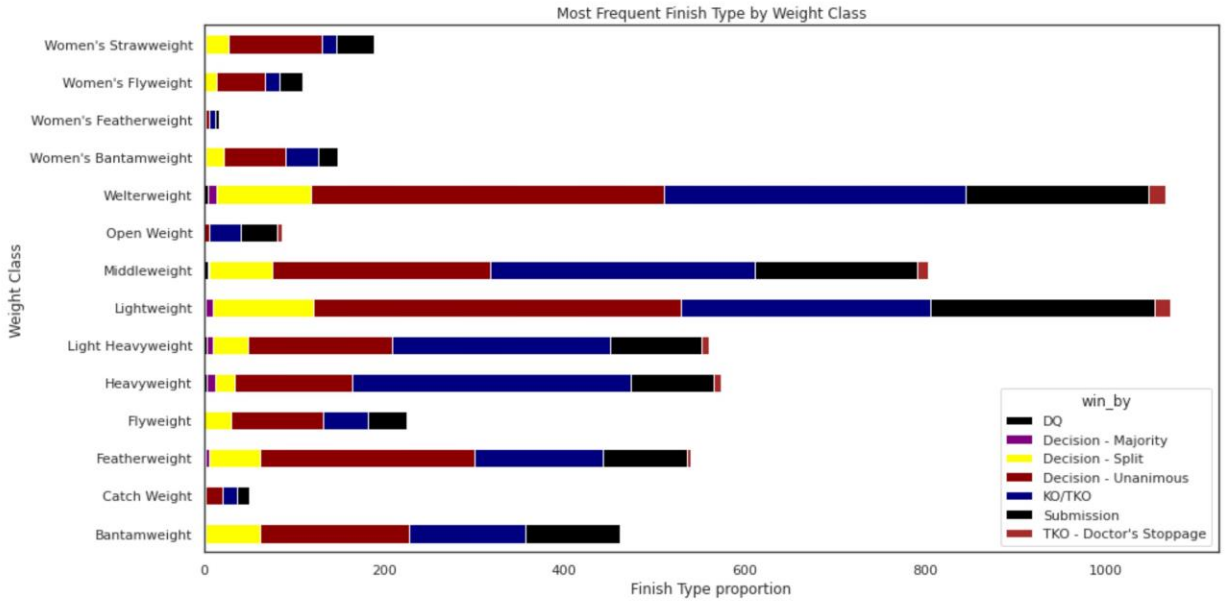
>_MONGOSH

```
> db.fighter.find({fighter_name:/F/})
< { _id: ObjectId("638eb49658e77d912eb65246"),
    Fighter_ID: 21,
    fighter_name: 'Fabio Aguiar',
    DOB: '10-Feb-88',
    SLpM: 0,
    Str_Acc: '0%',
    SApM: 0,
    Str_Def: '0%',
    TD_Avg: 0,
    TD_Acc: '0%',
    TD_Def: '0%',
    Sub_Avg: 0 }
  { _id: ObjectId("638eb49658e77d912eb652b1"),
    Fighter_ID: 128,
    fighter_name: 'Felipe Arantes',
    DOB: '9-Feb-88',
    SLpM: 2.37,
    Str_Acc: '45%',
    SApM: 2.66,
    Str_Def: '64%',
    TD_Avg: 0.93,
    TD_Acc: '47%',
    TD_Def: '41%',
    Sub_Avg: 0.7 }
```

DATA VISUALIZATION PYTHON

```
clss= ["black", "purple", "yellow", "darkred","navy", "black", "brown", "green", "pink"]

bar = df.groupby(['weight_class', 'win_by']).size().reset_index().pivot(columns='win_by', index='weight_class', values=0)
bar.plot(stacked=True, figsize=(15,8), kind="barh", color=clrs)
plt.xlabel('Finish Type proportion')
plt.ylabel('Weight Class')
plt.title('Most Frequent Finish Type by Weight Class');
```

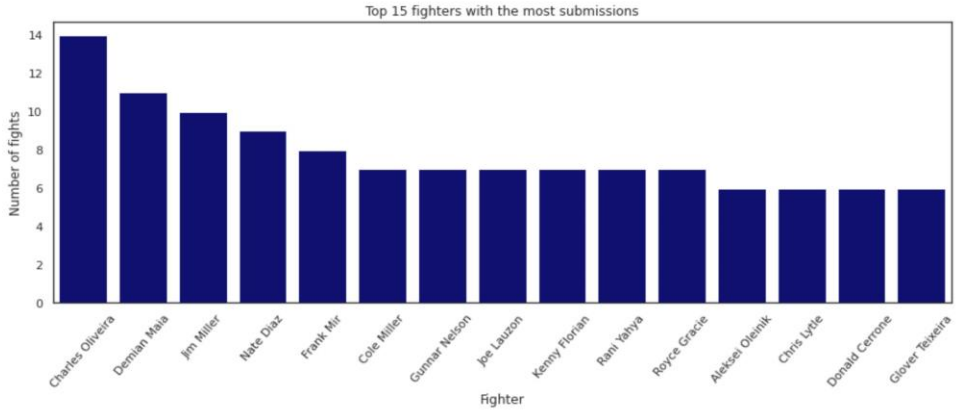


Who are the 15 fighters with the most submissions?

```
most_submissions_df = df.groupby('Winner')['is submission?'].agg('sum')
top_15_sub = most_submissions_df.nlargest(15)

index = top_15_sub.index
values = top_15_sub.values

plt.figure(figsize= (15,5))
sns.barplot(x=index, y=values, palette=clrs)
plt.xticks(rotation=50)
plt.xlabel('Fighter')
plt.ylabel('Number of fights')
plt.title('Top 15 fighters with the most submissions');
```



CONCLUSION

I think anyone who's interested in mixed martials arts would find valuable insights here. The intended audience can glean insights that help them think about where the future of the sport is going. It could be used by aspiring and current fighters to understand what fighting styles are important.

THANK YOU